

DAGUPAN, Philippines

WMO: 983250

Lat: 16.0868N

Lon: 120.3520E

Elev: 2

StdP: 101.30

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.1	21.9	17.6	12.6	22.9	18.7	13.6	23.1	10.1	25.7	8.4	27.1	1.9	180	0.354

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.3	34.8	28.3	34.0	28.2	33.3	28.0	29.7	32.8	29.2	32.3	28.8	32.0	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
29.0	25.6	32.1	28.5	24.9	31.7	28.0	24.1	31.3	98.7	32.9	96.1	32.3	94.0	32.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.0	6.0	18.5	37.4	2.0	1.2	17.0	38.3	15.9	39.0	14.7	39.6	13.3	40.5
			16.8	30.4	1.9	1.1	15.5	31.2	14.4	31.8	13.3	32.4	12.0	33.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	26.1	26.8	27.9	29.4	29.5	28.9	28.0	27.7	28.0	28.1	27.7	26.9	(6)
	DBStd	1.54	1.17	1.20	1.16	1.12	1.31	1.35	1.36	1.34	1.07	0.98	0.98	1.15	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	6541	501	469	556	582	604	566	558	548	540	562	532	523	(10)
	CDD18.3	3499	242	236	298	332	346	316	299	290	290	303	282	264	(11)
	CDH23.3	38170	2139	2264	3249	4113	4353	3678	3228	3006	3106	3376	3049	2608	(12)
(13)	CDH26.7	14114	598	729	1244	1878	2024	1504	1103	913	1029	1208	1064	819	(13)
(14)	Wind	WSAvg	2.5	2.5	2.7	2.9	3.0	2.7	2.4	2.3	2.5	2.8	2.1	2.2	(14)
(15) Precipitation	PrecAvg	2533	8	8	20	62	232	357	530	575	347	197	59	19	(15)
	PrecMax	4661	80	89	131	222	939	1122	2659	1500	946	807	358	136	(16)
	PrecMin	1256	0	0	0	0	53	63	12	18	31	7	0	0	(17)
	PrecStd	728	14	16	29	48	176	229	415	356	178	177	72	30	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	33.4	34.7	36.1	36.2	35.1	33.7	33.0	33.1	33.2	33.1	32.6	(19)
		MCWB	26.5	27.0	27.2	28.0	28.4	29.2	28.4	28.4	28.6	28.1	27.6	27.4	(20)
	2%	DB	31.1	32.2	33.5	34.8	34.7	34.0	32.7	32.0	32.1	32.3	32.2	31.6	(21)
		MCWB	26.2	26.7	27.2	28.0	28.5	28.8	28.1	28.1	28.0	27.8	27.4	27.0	(22)
	5%	DB	30.3	31.2	32.5	33.8	33.8	33.1	31.9	31.3	31.5	31.7	31.5	30.8	(23)
		MCWB	25.9	26.5	27.1	27.9	28.4	28.5	27.9	27.8	27.8	27.6	27.2	26.6	(24)
	10%	DB	29.5	30.3	31.5	32.9	32.9	32.2	31.1	30.6	30.8	31.1	30.8	30.1	(25)
		MCWB	25.6	26.1	26.8	27.7	28.2	28.1	27.6	27.4	27.5	27.4	27.1	26.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.2	29.1	29.5	30.2	30.4	30.4	29.5	29.5	29.6	29.4	29.4	28.7	(27)
		MCDB	30.7	31.3	32.5	33.6	34.0	33.6	32.4	31.8	31.9	31.8	31.3	31.1	(28)
	2%	WB	27.2	28.0	28.5	29.2	29.5	29.5	28.7	28.5	28.6	28.6	28.4	27.9	(29)
		MCDB	30.0	30.9	31.8	32.7	33.1	32.8	31.6	31.2	31.2	31.3	30.9	30.5	(30)
	5%	WB	26.6	27.2	27.7	28.6	28.9	28.9	28.1	28.0	28.1	28.1	27.7	27.2	(31)
		MCDB	29.5	30.2	31.2	32.3	32.6	32.3	31.3	30.8	30.8	31.0	30.7	30.1	(32)
	10%	WB	26.0	26.5	27.1	28.1	28.3	28.3	27.6	27.5	27.6	27.6	27.2	26.6	(33)
		MCDB	29.0	29.6	30.6	32.0	32.1	31.6	30.6	30.1	30.3	30.5	30.3	29.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	7.6	7.8	7.9	7.3	6.6	5.7	4.9	5.6	6.2	6.6	6.8	(35)
		MCDBR	8.3	9.0	9.4	9.2	8.6	7.9	7.1	6.5	6.7	7.2	7.6	7.8	(36)
	5% WB	MCWBR	5.2	5.5	5.1	4.4	4.2	4.1	3.7	3.6	3.7	3.9	4.2	4.8	(37)
		MCWBR	7.7	8.2	8.3	8.2	7.9	7.6	6.8	6.1	6.4	6.8	7.1	7.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.405	0.419	0.468	0.520	0.488	0.470	0.470	0.476	0.472	0.481	0.425	0.415	(40)
		taud	2.430	2.379	2.254	2.120	2.246	2.312	2.313	2.286	2.287	2.243	2.419	2.428	(41)
	Ebn at Noon		884	888	851	804	819	825	827	829	835	819	858	863	(42)
		Edh at Noon	111	122	142	162	141	131	131	136	135	138	112	109	(43)
(44) All-Sky Solar Radiation	RadAvg		4.76	5.42	6.00	6.38	5.76	5.26	4.70	4.41	4.77	4.89	4.58	4.34	(44)
	RadStd		0.20	0.33	0.26	0.28	0.45	0.60	0.62	0.48	0.42	0.38	0.39	0.27	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (13 neighbors)	+0.38	N/A	N/A	+0.40	+0.71	N/A	N/A	N/A	N/A	N/A
		+0.32	+0.57	N/A	+0.40	+0.36	+0.42	N/A	N/A	+158	+158

Nomenclature: See separate page